Econometrics Student Experiences the Machine Learning Framing of Regression, For Once in His Life

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Abstract

Inferential vs predictive notions of the same statistical process: linear regression.

1 Inference vs prediction in statistics

1.1 In econometrics we infer

- In econometrics, regression is the engine of **inference**.
- Inference, in this sense, means we use existing data to infer the *relationship* between some variables (e.g. $x =$ income and $y =$ SAT score).
- To be more precise, this relationship is a statistical distribution. Our goal is to model that distribution.
- When we do regression, we assume that there's a *true* distribution; a **data generating process**.
- When we do linear regression, we assume that the true data generating process looks like a multivariate linear equation:
$$y = X\beta + \epsilon$$
- Our **object of interest** is the vector of weights: β
- Put differently, β is the vector of true relationships between our variables. ϵ is the noise.
- Our goal is to estimate the weights of the vector β which lead to the most accurate predictions for y:
$$\hat{y} = X\hat{\beta}$$

1.2 In ML we predict

- In machine learning, our primary goal is not to obtain and report a relationship (like β), but rather to use data from the world to *make more data*.
- In other words, we take a complex set of causal relationships out there in the world—samples from a data generating process—and use it to **predict** what that process would output.

- LLMs don't explicitly mimick the causal process in your brain which leads to language (emotional disregulation, excitement, boredom, etc); they find complex statistical *relationships* between tokens (samples from the process) and predict what the process would output (the next word).
- Thus, our object of interest is not β, but y.
- Instead of focusing on obtaining and reporting $\hat{\beta}$, we do so for $\hat{y}$.

1.3 Throw ϵ out the window (for now)

- FINISH THIS

2 Linear regression is just projecting onto span(X)

3 Gradient descent for OLS

4 OLS is MLE under Gaussian noise

5 Logistic regression doesn't have closed form

6 Regularizing is biasing on purpose